Newton-Raphson Method

- Newton's (or the Newton-Raphson) method is one of the most powerful and well-known numerical methods for solving a root-finding problem.

- Given an initial guess of the root x_0 , Newton-Raphson method uses information about the function and its derivative at that point to find a better guess of the root.

Assumptions:

- $f(x)$ is continuous and first derivative is known
- An initial guess x_0 such that $f'(x_0) \neq 0$ is given

Derivation of Newton's Method

- Let x_0 be an approximation of the root of $f(x) = 0$, whose real root is p. Thus, $p = x_0 + h$, where h is the correction (small) to be applied to x_0 to give the exact value of the root. Therefore,

$$f(p) = f(x_0 + h) = 0$$

- By Taylor series expansion we get,

$$f(x_0) + hf'(x_0) + \frac{h^2}{2!} f''(x_0) + \cdots = 0$$

- Since h is small, neglecting higher orders of h we get,

$$\boxed{f(x_0) + hf'(x_0) \approx 0 \quad \Rightarrow h = -\frac{f(x_0)}{f'(x_0)}}$$

Derivation of Newton's Method

- Substituting this value of h in $p = x_0+h$ we get a better approximation to the root of $f(x) = 0$ as

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

- Therefore, the successive approximations are

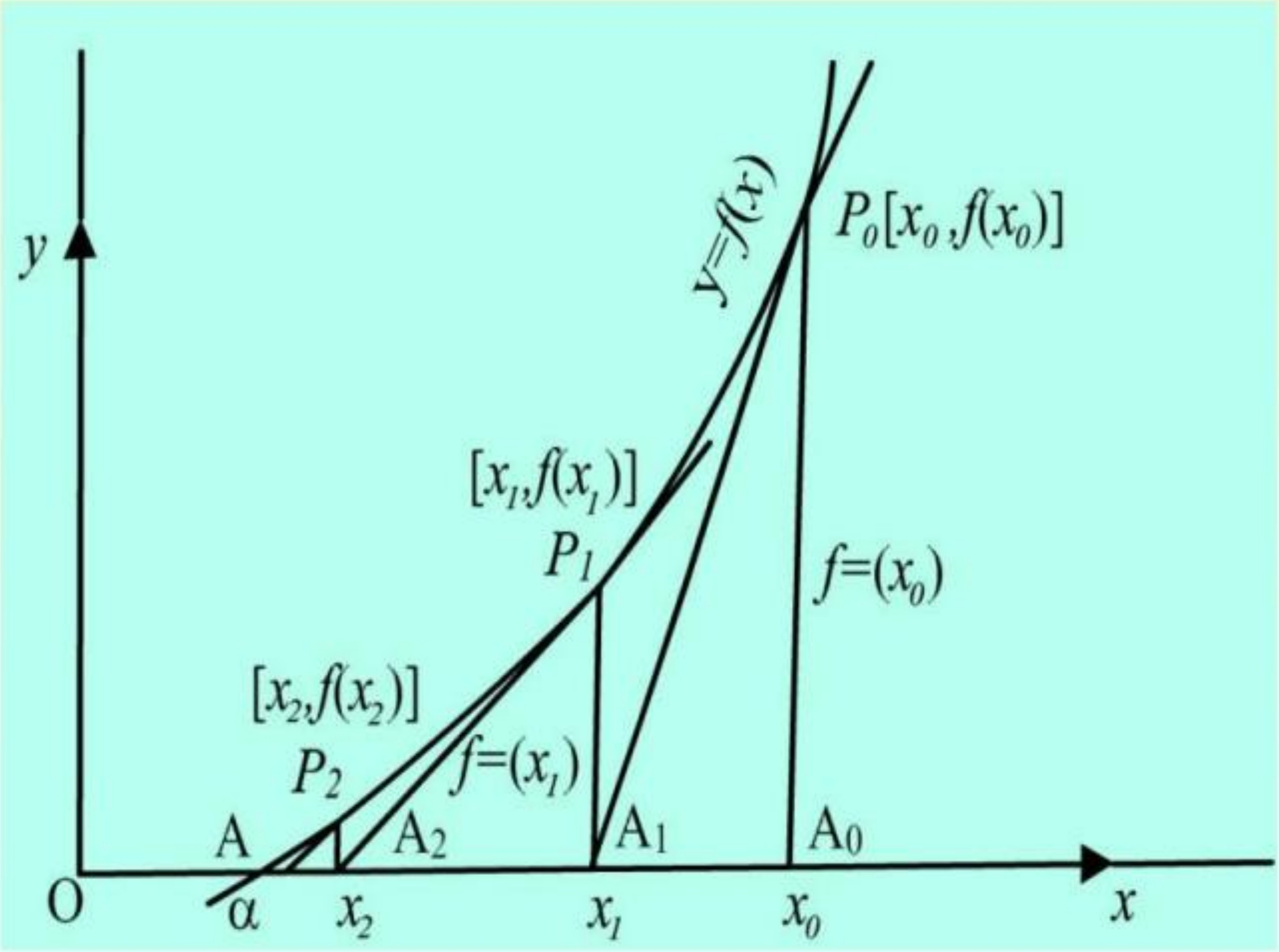
$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\cdots\cdots\cdots\cdots\cdots\cdots\cdots$$

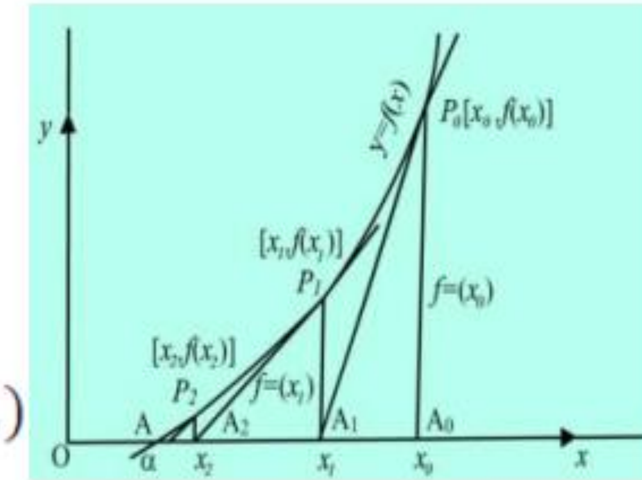
$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

- This formula is known as the iteration formula for Newton Raphson Method

Geometrical significance



The figure represents a magnified view of the graph $y = f(x)$ where it crosses the X-axis at $x = \alpha$. Let x_o be the initial approximation at the starting point. To calculate the first approximation we replace the graph at $P_0(x_0, y_o = f(x_o))$ by the tangent at P_o. This tangent will intersect the x-axis at some point A_1 whose abscissa is $OA_1 = x_1$.



Then, from $\Delta P_o A_1 A_o = \frac{PoAo}{A_1 A_o} = \frac{f(x_0)}{x_0 - x_1} = \frac{yo}{x_0 - x_1}$

But $\tan P_0 A_1 A_o$= Slope of the tangent at P_o=$f'(x_o)$

Therefore,

$$f'(x_0) = \frac{f(x_o)}{x_o - x_1} \Rightarrow x_1 = x_o - \frac{f(x_o)}{f'(x_o)}$$

Using x_1 as starting point, the tangent at $P_1[x_1, f(x_1)]$ will give $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

Hence OA_1, OA_2 , … are successive approximations to the desired root.

Example

- Find by Newton-Raphson Method the real root of

$$3x - \cos x - 1 = 0$$

- Taking x0 = 0.5, the successive approximations of the root are computed in the following table:

n	x_n	$f(x_n)$	$f'(x_n)$	$h_n = -\frac{f(x_n)}{f'(x_n)}$	$x_{n+1} = x_n + h_n$
0	0.5	-0.37	3.48	0.1063	0.6063
1	0.6063	-0.00286	3.56983	0.000801	0.607101
2	0.607101	-0.00000231	3.570489	0.00000064	0.60710164
3	0.60710106	-0.00000017	3.570489	0.00000003	0.60710163

Example 9: Find a real root of $x^x + x - 4 = 0$, by Newton-Raphson method, correct to six decimal places.

Taking x_0 = 1.6, the successive iterations are computed in the following table:

n	x_n	$f(x_n)$	$f'(x_n)$	$h_n = -\frac{f(x)}{f'(x)}$	$x_{n+1} = x_n + h_n$
0	1.6	-0.27	4.12	0.066	1.666
1	1.666	0.0065	4.5352	-0.00143	1.66457
2	1.66457	0.0000318	4.525536	-0.000007	1.664563
3	1.664563	0.0000002	4.5254887	-0.00000004	1.664563

Thus, 1.664563 is a root of $f(x) = 0$, correct up to six decimal places.

Example 10: Find the cube of 10, that is, $\sqrt[3]{10}$, correct to 4- significant figures.

Solution. Let $x=\sqrt[3]{10}$, then $x^3-10=0$. Let $f(x)=x^3-10 \Rightarrow f'(x)=3x^2$. Hence Newton Raphson's iterative formula gives

$$x_{n+1}=x_n-\frac{x_n^3-10}{3x_n^2}=\frac{2x_n^3+10}{3x_n^2} \Rightarrow x_{n+1}=\frac{1}{3}\left(2x_n+\frac{10}{x_n^2}\right)$$

Since $8^{1/3}=2$ and $27^{1/3}=3$, we may take initially $x_n = 2.2$. Therefore,

$$x_1=\frac{1}{3}\left(2x_0+\frac{10}{x_0^2}\right)=\frac{1}{3}\left(2\times 2.2+\frac{10}{(2.2)^2}\right)=2.15537$$

$$x_2=\frac{1}{3}\left(2\times 2.15537+\frac{10}{(2.15537)^2}\right)=2.15444$$

$$x_3=\frac{1}{3}\left(2\times 2.15444+\frac{10}{(2.15444)^2}\right)=2.15443$$

$$x_4=\frac{1}{3}\left(2\times 2.15443+\frac{10}{(2.15443)^2}\right)=2.15443$$

Therefore, $\sqrt[3]{10}=2.154$, correct to 4-significant figures.

Example 11: Find a real root of $\log_e(x) = \cos(x)$ by Newton Raphson method correct to 6-significant figures.

Exercise 7: Find a real root of $\sinh(x) - x = 0$ by Newton Raphson method, correct to three significant figures. (Ans. 0.100)

Exercise 8: Find a real root of $\tan(x) - \tanh(x) = 0$ by Newton Raphson method, correct to three significant figures. (Ans. 0.619)